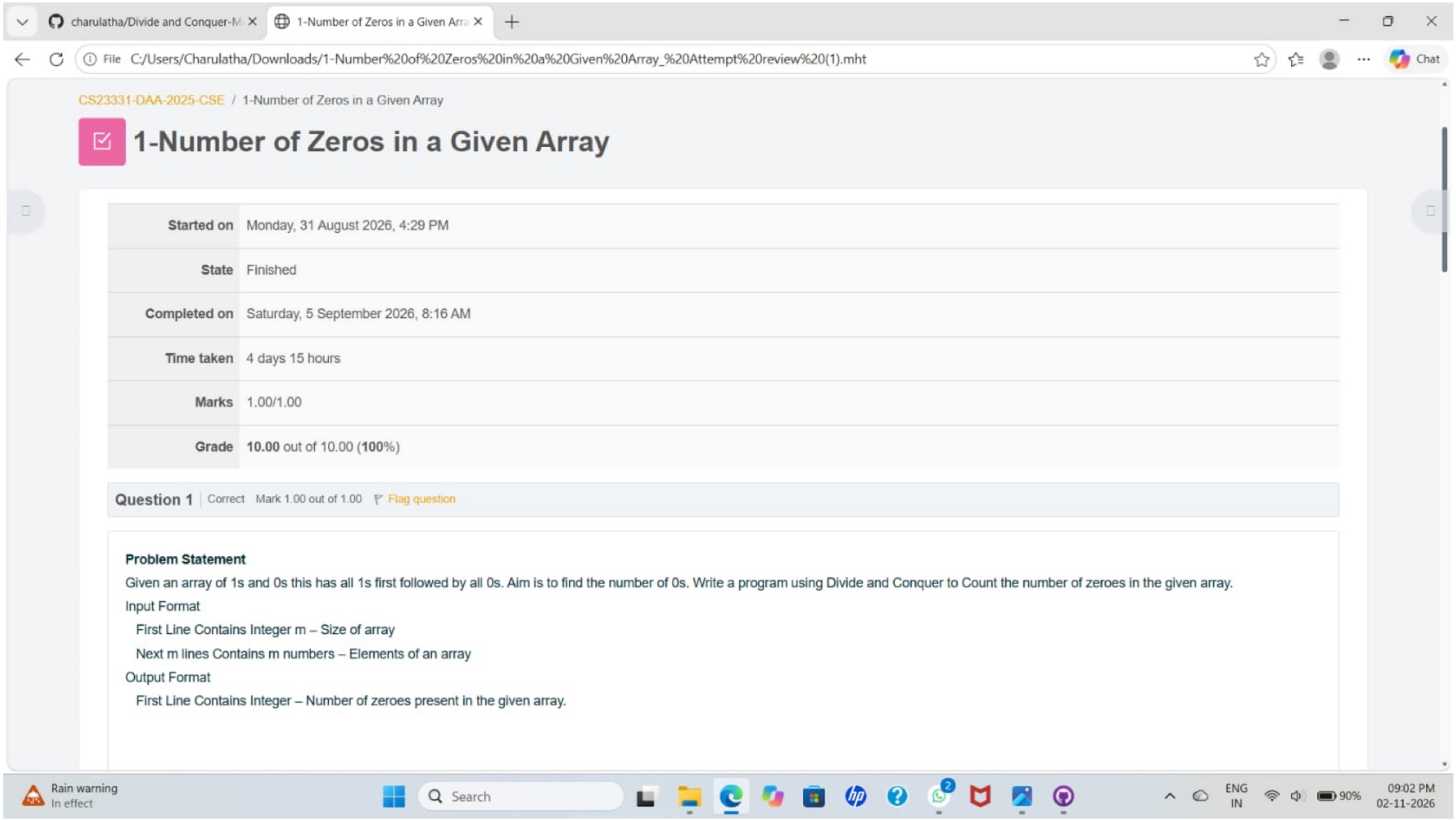
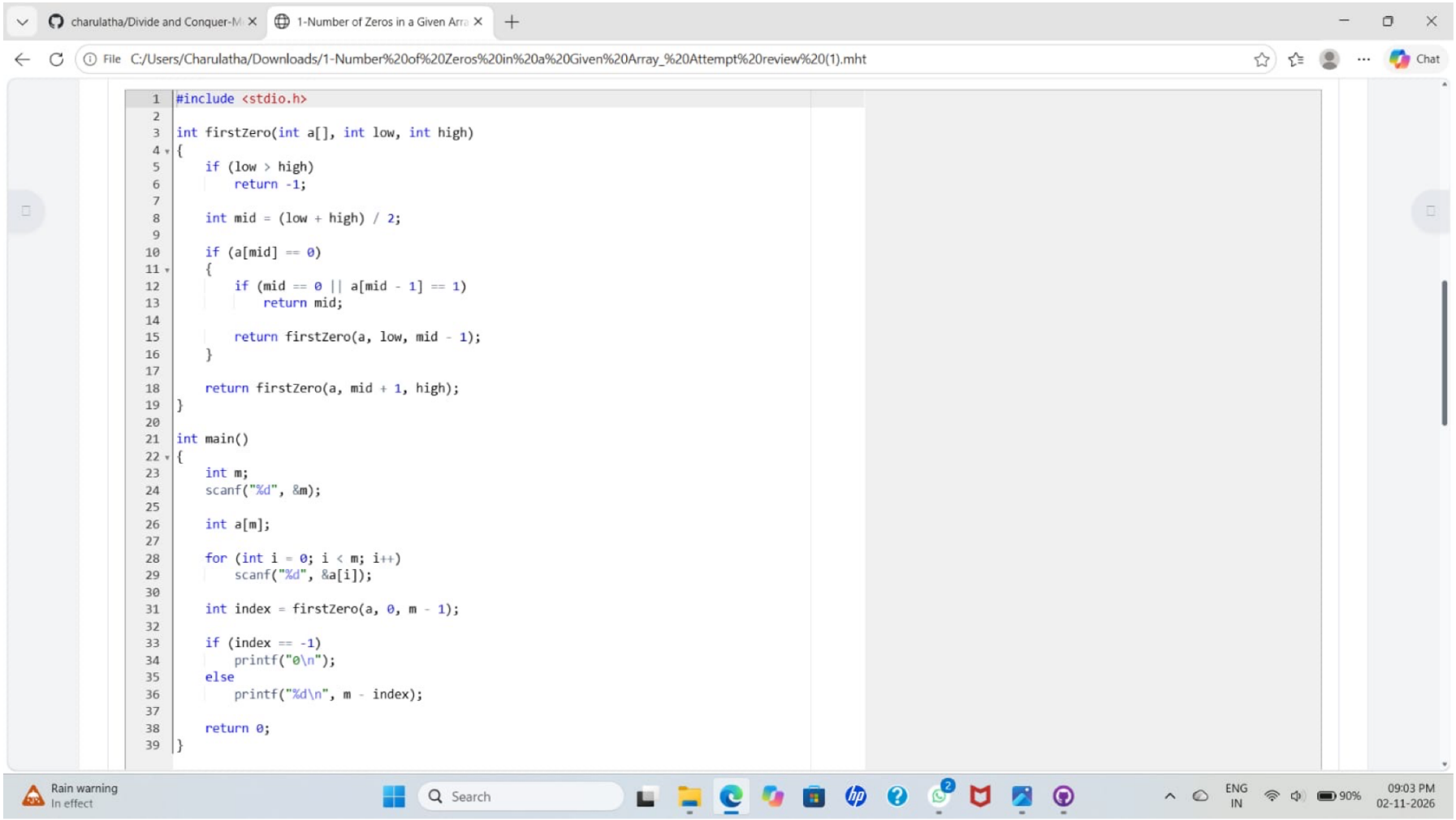
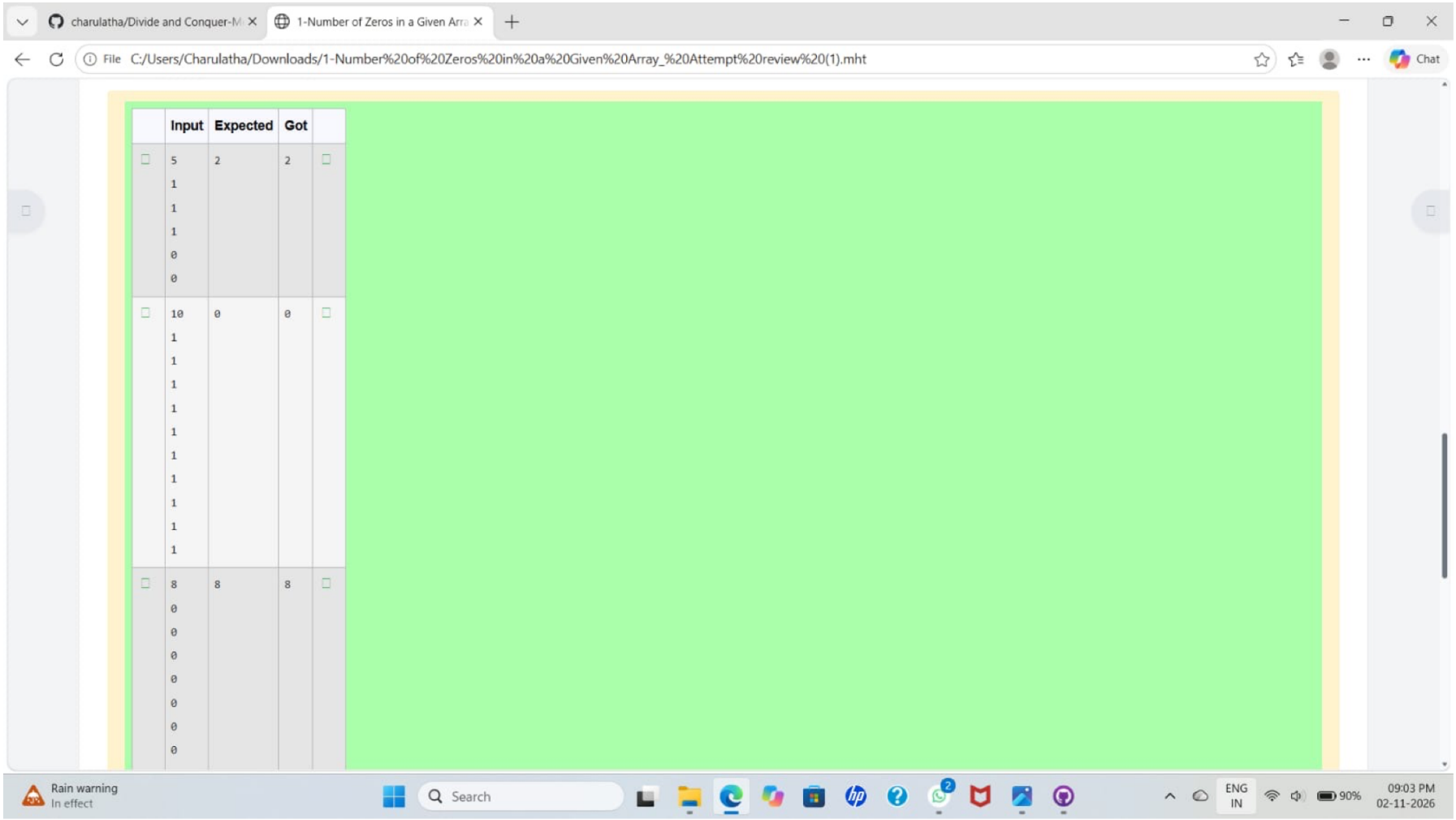
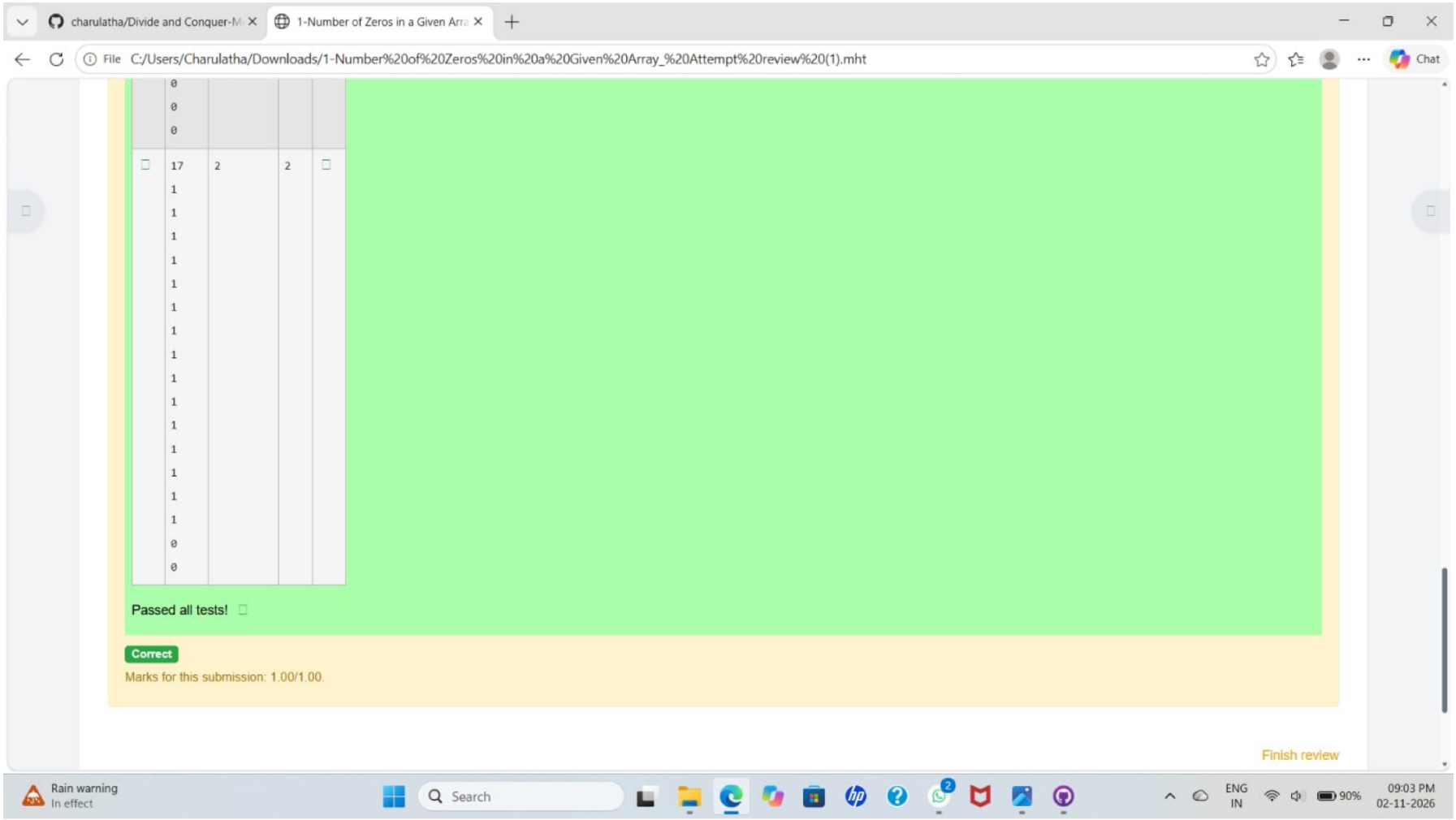
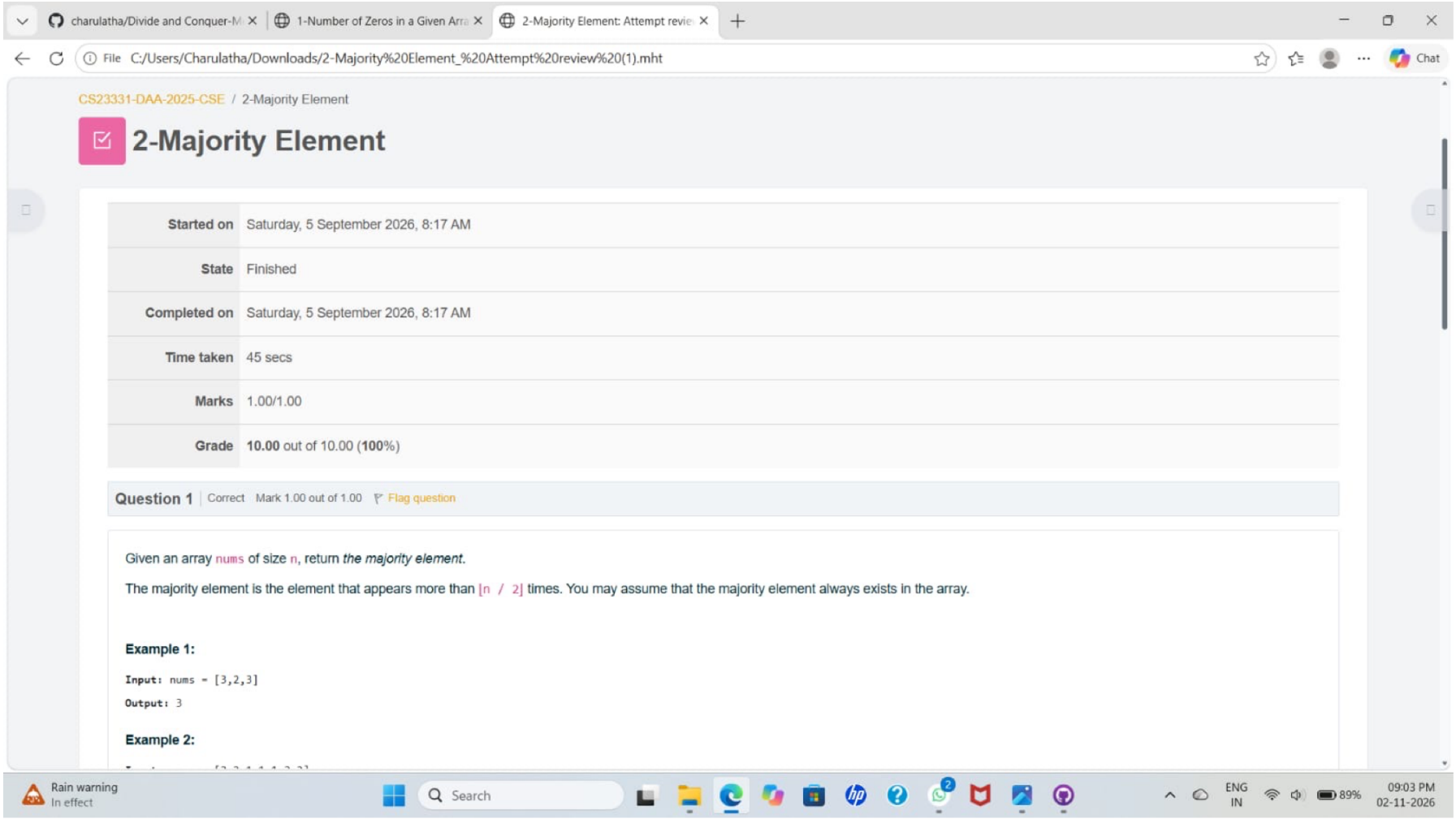
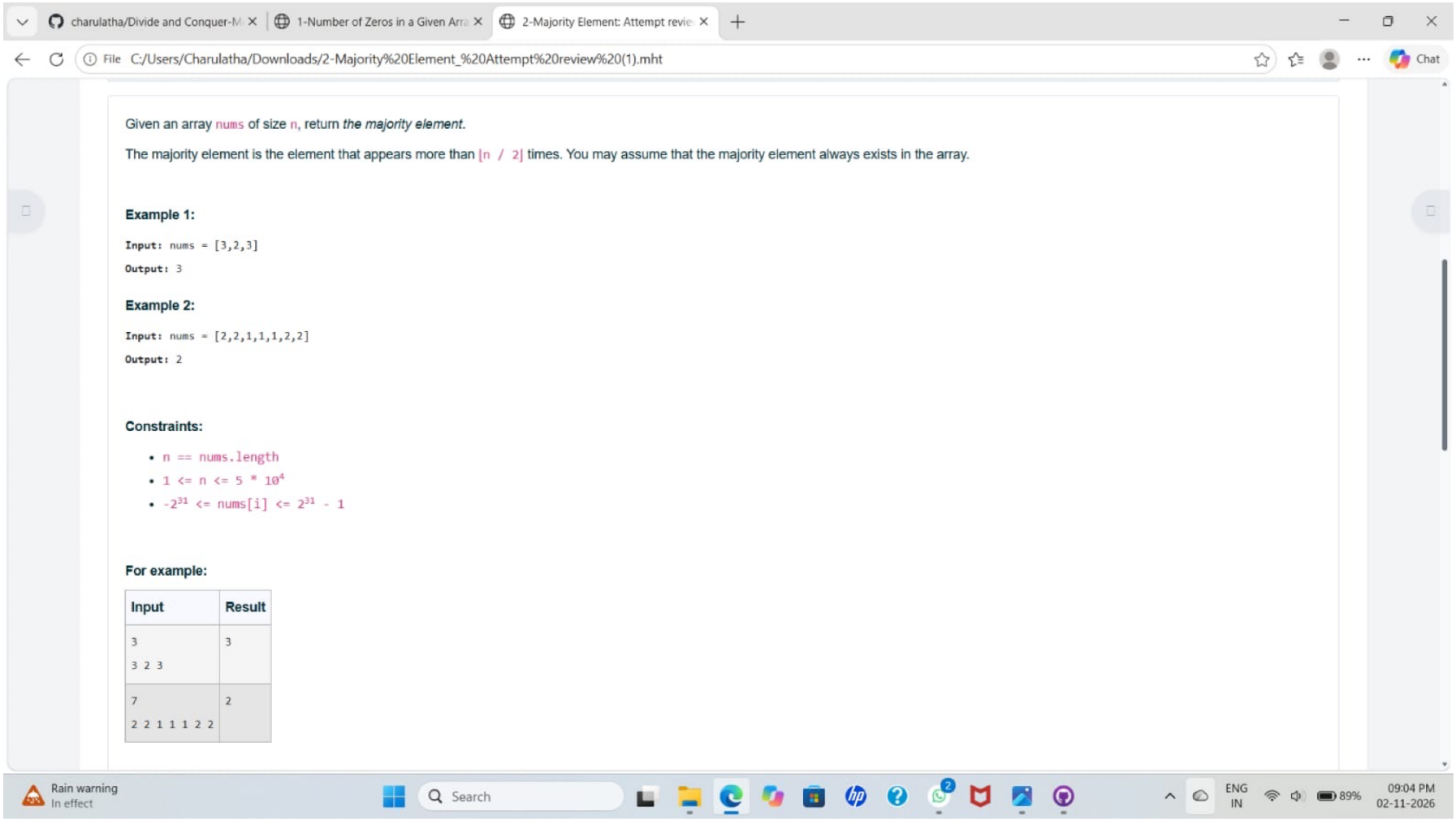

charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
File C:/Users/Charulatha/Downloads/1-Number%20of%20Zeros%20in%20a%20Given%20Array_%20Attempt%20review%20(1).mht
Chat
CS23331-DAA-2025-CSE / 1-Number of Zeros in a Given Array
1-Number of Zeros in a Given Array
Started on Monday, 31 August 2026, 4:29 PM
State Finished
Completed on Saturday, 5 September 2026, 8:16 AM
Time taken 4 days 15 hours
Marks 1.00/1.00
Grade 10.00 out of 10.00 (100%)
Question 1 Correct Mark 1.00 out of 1.00 Flag question
Problem Statement
Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.
Input Format
First Line Contains Integer m – Size of array
Next m lines Contains m numbers – Elements of an array
Output Format
First Line Contains Integer – Number of zeroes present in the given array.
Rain warning
In effect
Search
ENG
IN
90%
09:02 PM
02-11-2026

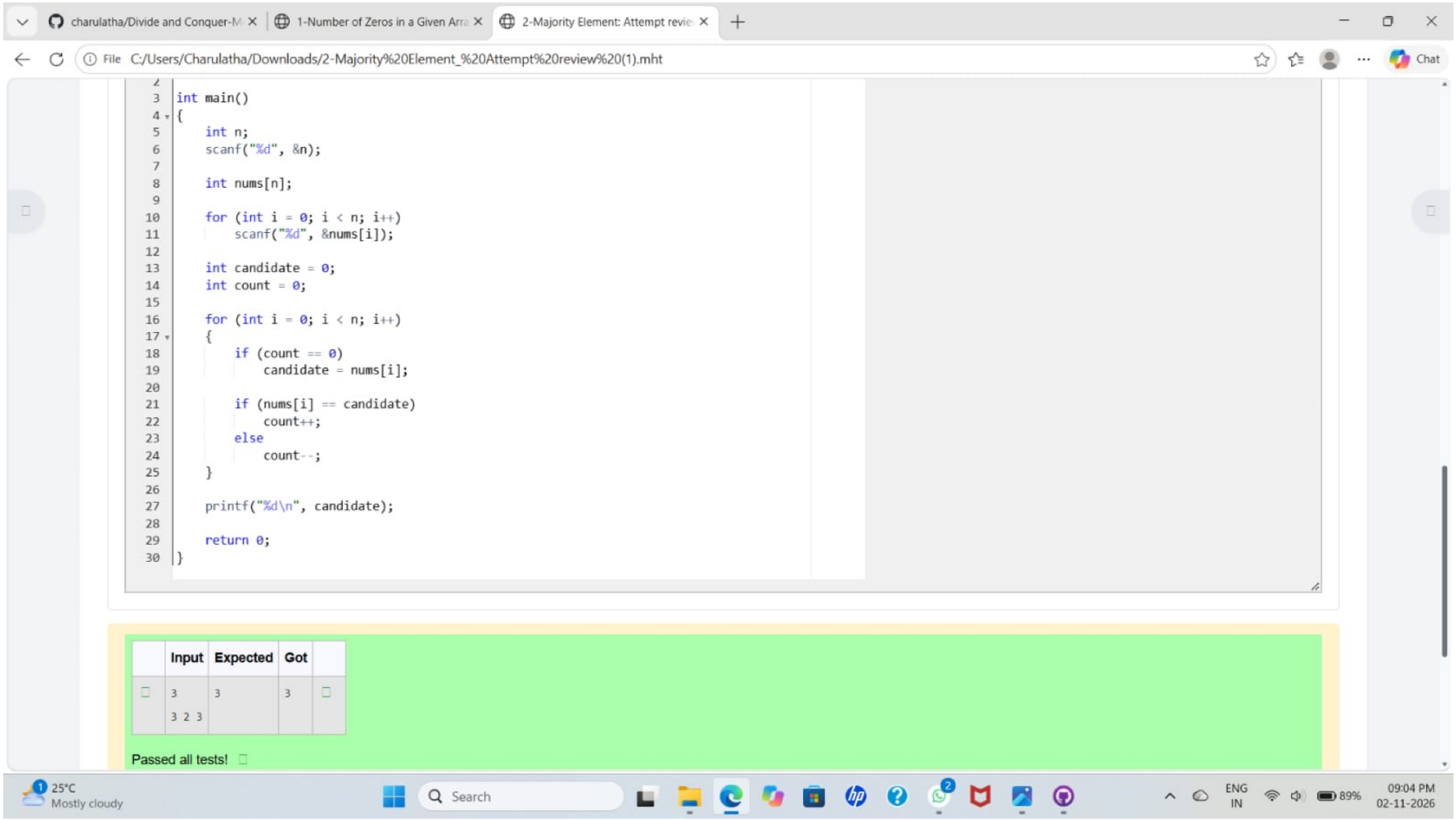

charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
File C:/Users/Charulatha/Downloads/1-Number%20of%20Zeros%20in%20a%20Given%20Array_%20Attempt%20review%20(1).mht
Chat
#include <stdio.h>
int firstZero(int a[], int low, int high)
{
if (low > high)
return -1;
int mid = (low + high) / 2;
if (a[mid] == 0)
{
if (mid == 0 || a[mid - 1] == 1)
return mid;
return firstZero(a, low, mid - 1);
}
return firstZero(a, mid + 1, high);
}
int main()
{
int m;
scanf("%d", &m);
int a[m];
for (int i = 0; i < m; i++)
scanf("%d", &a[i]);
int index = firstZero(a, 0, m - 1);
if (index == -1)
printf("0\n");
else
printf("%d\n", m - index);
return 0;
}
Rain warning
In effect
Search
ENG
IN
90%
09:03 PM
02-11-2026

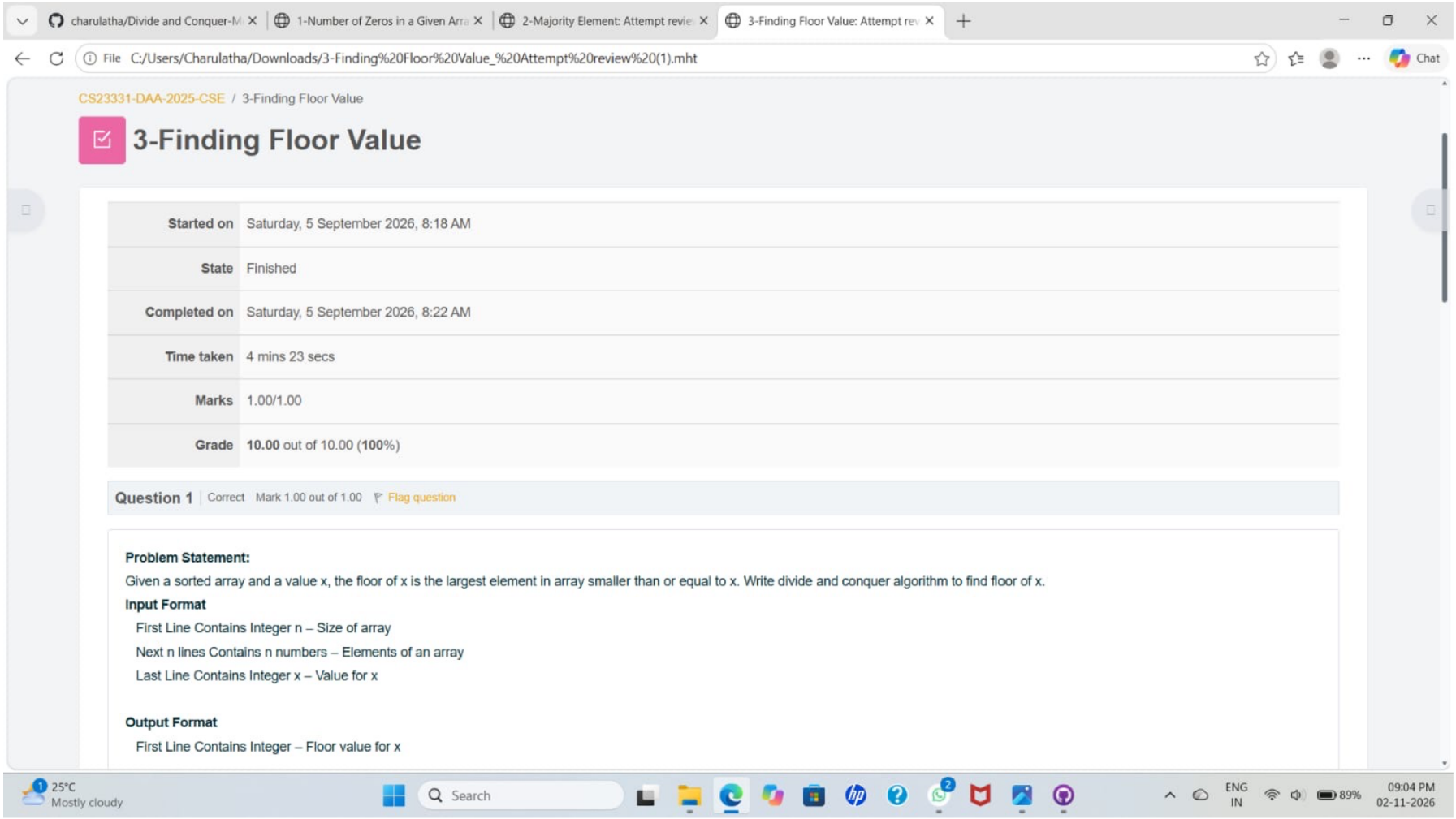

charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
File C:/Users/Charulatha/Downloads/1-Number%20of%20Zeros%20in%20a%20Given%20Array_%20Attempt%20review%20(1).mht
Chat
Input
Expected
Got
5 1 1 1 0 0
2
2
10 1 1 1 1 1 1 1 1 1
0
0
8 0 0 0 0 0 0 0
8
8
Rain warning
In effect
Search
ENG
IN
90%
09:03 PM
02-11-2026

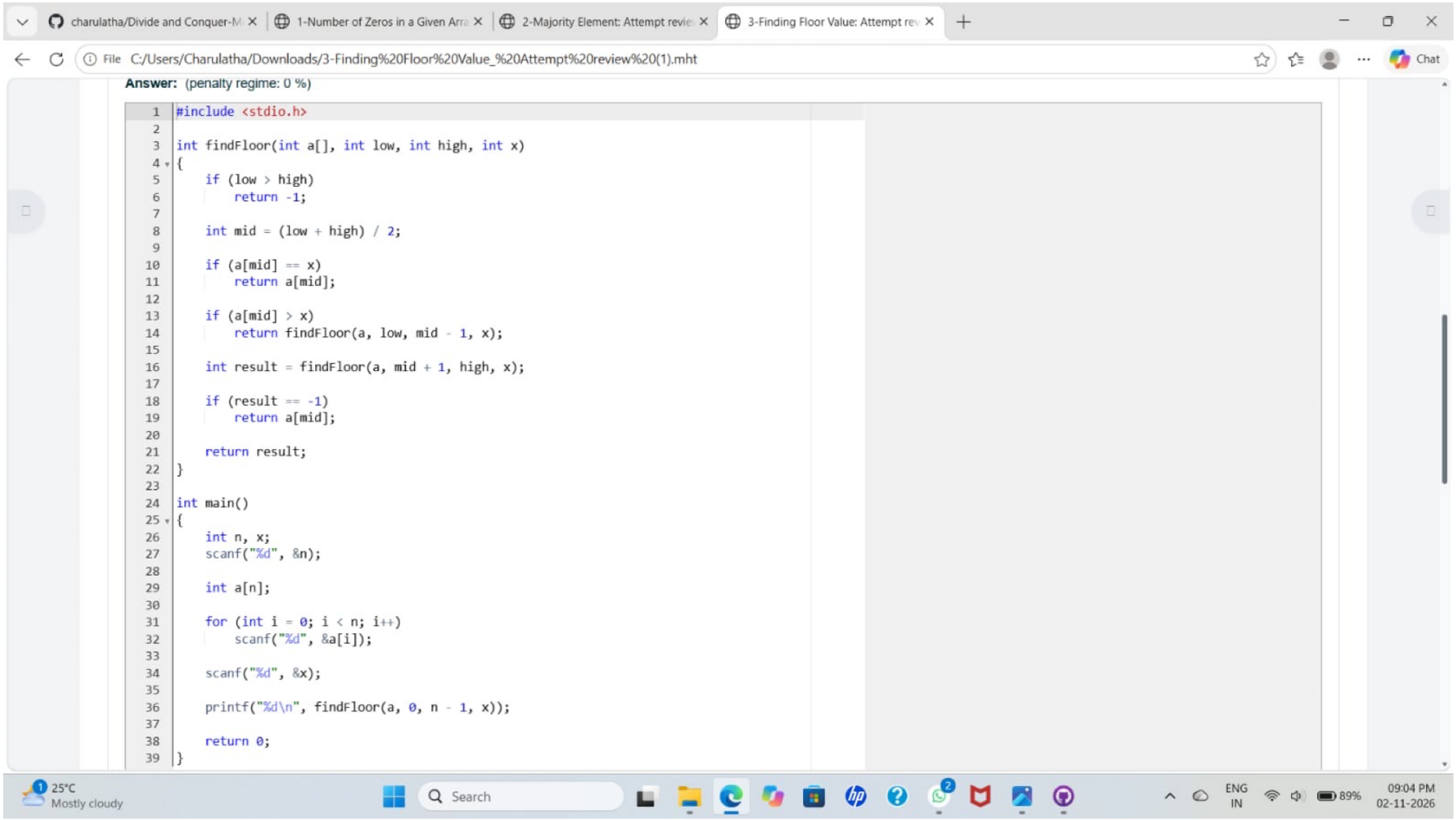

charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
File C:/Users/Charulatha/Downloads/1-Number%20of%20Zeros%20in%20a%20Given%20Array_%20Attempt%20review%20(1).mht
Chat
0
0
0
17
1
1
1
1
1
1
1
1
1
1
1
1
1
1
1
1
1
0
0
2
2
Passed all tests!
Correct
Marks for this submission: 1.00/1.00.
Finish review
Rain warning
In effect
Search
ENG
IN
90%
09:03 PM
02-11-2026

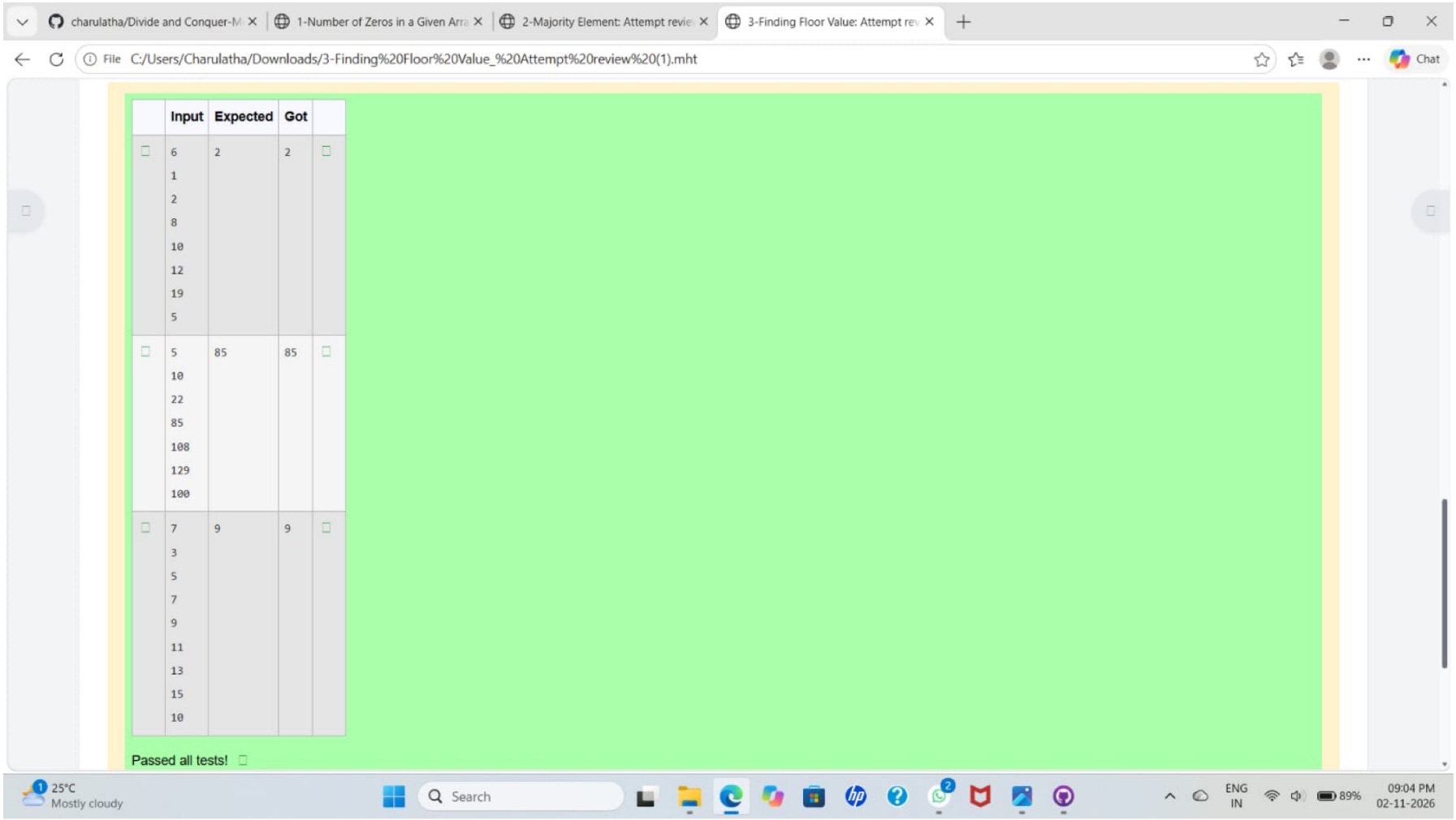

charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
2-Majority Element: Attempt revie
File C:/Users/Charulatha/Downloads/2-Majority%20Element_%20Attempt%20review%20(1).mht
Chat
CS23331-DAA-2025-CSE / 2-Majority Element
2-Majority Element
Started on Saturday, 5 September 2026, 8:17 AM
State Finished
Completed on Saturday, 5 September 2026, 8:17 AM
Time taken 45 secs
Marks 1.00/1.00
Grade 10.00 out of 10.00 (100%)
Question 1 Correct Mark 1.00 out of 1.00 Flag question
Given an array nums of size n, return the majority element.
The majority element is the element that appears more than ⌊n / 2⌋ times. You may assume that the majority element always exists in the array.
Example 1:
Input: nums = [3,2,3]
Output: 3
Example 2:
Rain warning
In effect
Search
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02-11-2026

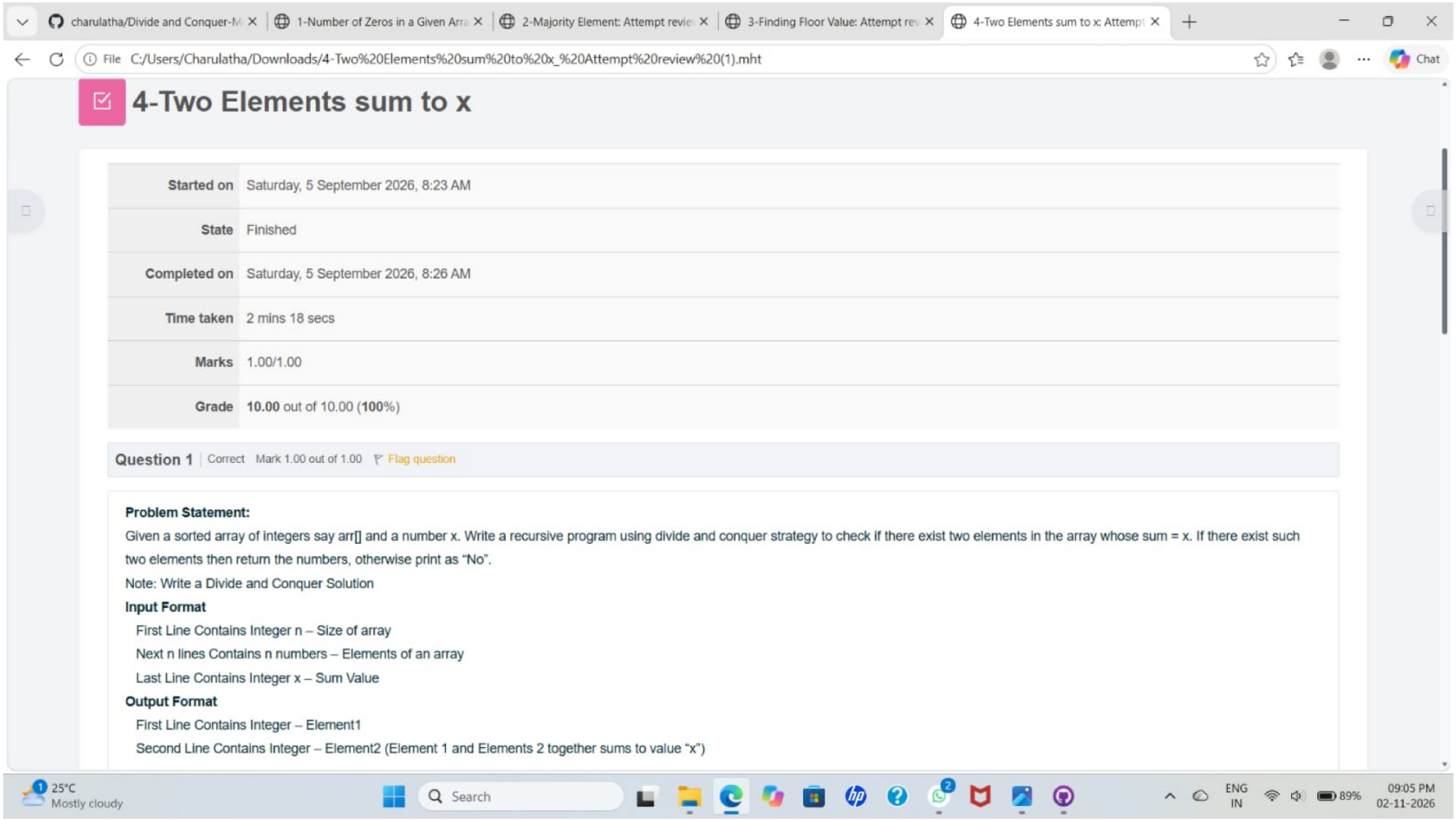

charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
2-Majority Element: Attempt revie
File C:/Users/Charulatha/Downloads/2-Majority%20Element_%20Attempt%20review%20(1).mht
Chat
Given an array nums of size n, return the majority element.
The majority element is the element that appears more than ⌊n / 2⌋ times. You may assume that the majority element always exists in the array.
Example 1:
Input: nums = [3,2,3]
Output: 3
Example 2:
Input: nums = [2,2,1,1,1,2,2]
Output: 2
Constraints:
n == nums.length
1 <= n <= 5 * 10^4
-2^31 <= nums[i] <= 2^31 - 1
For example:
Input
Result
3
3 2 3
3
7
2 2 1 1 1 2 2
2
Rain warning
In effect
Search
ENG
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89%
09:04 PM
02-11-2026


charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
2-Majority Element: Attempt revie
File C:/Users/Charulatha/Downloads/2-Majority%20Element_%20Attempt%20review%20(1).mht
Chat
int main()
{
int n;
scanf("%d", &n);
int nums[n];
for (int i = 0; i < n; i++)
scanf("%d", &nums[i]);
int candidate = 0;
int count = 0;
for (int i = 0; i < n; i++)
{
if (count == 0)
candidate = nums[i];
if (nums[i] == candidate)
count++;
else
count--;
}
printf("%d\n", candidate);
return 0;
}
Input
Expected
Got
3
3 2 3
3
3
Passed all tests!
25°C
Mostly cloudy
Search
ENG
IN
89%
09:04 PM
02-11-2026


charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
2-Majority Element: Attempt revie
3-Finding Floor Value: Attempt rev
File C:/Users/Charulatha/Downloads/3-Finding%20Floor%20Value_%20Attempt%20review%20(1).mht
Chat
CS23331-DAA-2025-CSE / 3-Finding Floor Value
3-Finding Floor Value
Started on Saturday, 5 September 2026, 8:18 AM
State Finished
Completed on Saturday, 5 September 2026, 8:22 AM
Time taken 4 mins 23 secs
Marks 1.00/1.00
Grade 10.00 out of 10.00 (100%)
Question 1 Correct Mark 1.00 out of 1.00 Flag question
Problem Statement:
Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.
Input Format
First Line Contains Integer n – Size of array
Next n lines Contains n numbers – Elements of an array
Last Line Contains Integer x – Value for x
Output Format
First Line Contains Integer – Floor value for x
25°C
Mostly cloudy
Search
ENG
IN
89%
09:04 PM
02-11-2026


charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
2-Majority Element: Attempt revie
3-Finding Floor Value: Attempt rev
File C:/Users/Charulatha/Downloads/3-Finding%20Floor%20Value_%20Attempt%20review%20(1).mht
Chat
Answer: (penalty regime: 0 %)
#include <stdio.h>
int findFloor(int a[], int low, int high, int x)
{
if (low > high)
return -1;
int mid = (low + high) / 2;
if (a[mid] == x)
return a[mid];
if (a[mid] > x)
return findFloor(a, low, mid - 1, x);
int result = findFloor(a, mid + 1, high, x);
if (result == -1)
return a[mid];
return result;
}
int main()
{
int n, x;
scanf("%d", &n);
int a[n];
for (int i = 0; i < n; i++)
scanf("%d", &a[i]);
scanf("%d", &x);
printf("%d\n", findFloor(a, 0, n - 1, x));
return 0;
}
25°C
Mostly cloudy
Search
ENG
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09:04 PM
02-11-2026


charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
2-Majority Element: Attempt revie
3-Finding Floor Value: Attempt rev
File C:/Users/Charulatha/Downloads/3-Finding%20Floor%20Value_%20Attempt%20review%20(1).mht
Chat
Input
Expected
Got
6 1 2 8 10 12 19 5
2
2
5 10 22 85 108 129 100
85
85
7 3 5 7 9 11 13 15 10
9
9
Passed all tests!
25°C Mostly cloudy
Search
ENG IN
89%
09:04 PM 02-11-2026


charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
2-Majority Element: Attempt revie
3-Finding Floor Value: Attempt rev
4-Two Elements sum to x: Attempt
File C:/Users/Charulatha/Downloads/4-Two%20Elements%20sum%20to%20x_%20Attempt%20review%20(1).mht
Chat
4-Two Elements sum to x
Started on Saturday, 5 September 2026, 8:23 AM
State Finished
Completed on Saturday, 5 September 2026, 8:26 AM
Time taken 2 mins 18 secs
Marks 1.00/1.00
Grade 10.00 out of 10.00 (100%)
Question 1 Correct Mark 1.00 out of 1.00 Flag question
Problem Statement:
Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".
Note: Write a Divide and Conquer Solution
Input Format
First Line Contains Integer n – Size of array
Next n lines Contains n numbers – Elements of an array
Last Line Contains Integer x – Sum Value
Output Format
First Line Contains Integer – Element1
Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")
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Mostly cloudy
Search
ENG
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89%
09:05 PM
02-11-2026

charulatha/Divide and Conquer-M | 1-Number of Zeros in a Given Arra | 2-Majority Element: Attempt revie | 3-Finding Floor Value: Attempt rev | 4-Two Elements sum to x: Attempt

File C:/Users/Charulatha/Downloads/4-Two%20Elements%20sum%20to%20x_%20Attempt%20review%20(1).mht

Chat

```c
#include <stdio.h>
int findPair(int a[], int left, int right, int x, int *e1, int *e2)
{
    if (left >= right)
        return 0;
    int sum = a[left] + a[right];
    if (sum == x)
    {
        *e1 = a[left];
        *e2 = a[right];
        return 1;
    }
    if (sum > x)
        return findPair(a, left, right - 1, x, e1, e2);
    return findPair(a, left + 1, right, x, e1, e2);
}
int main()
{
    int n, x;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    int e1, e2;

    if (findPair(a, 0, n - 1, x, &e1, &e2))
    {
        printf("%d\n", e1);
        printf("%d\n", e2);
    }
    else
    {
        printf("No\n");
    }

    return 0;
}
```

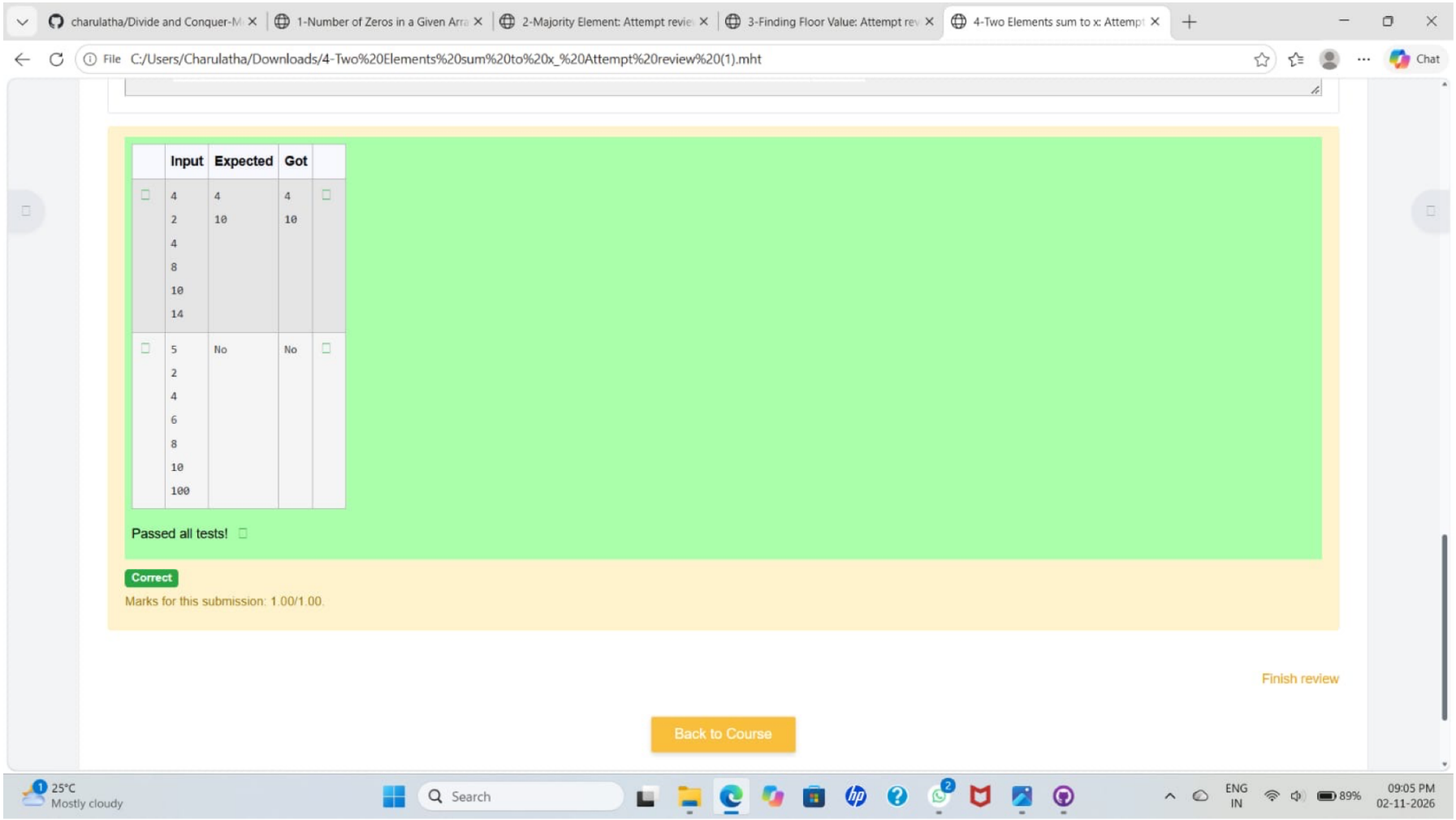
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Search

ENG IN

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charulatha/Divide and Conquer-M
1-Number of Zeros in a Given Arra
2-Majority Element: Attempt revie
3-Finding Floor Value: Attempt rev
4-Two Elements sum to x: Attempt
File C:/Users/Charulatha/Downloads/4-Two%20Elements%20sum%20to%20x_%20Attempt%20review%20(1).mht
Chat
Input
Expected
Got
4
2
4
8
10
14
4
10
4
10
5
2
4
6
8
10
100
No
No
Passed all tests!
Correct
Marks for this submission: 1.00/1.00.
Finish review
Back to Course
25°C
Mostly cloudy
Search
ENG
IN
89%
09:05 PM
02-11-2026

charulatha/Divide and Conqu | 1-Number of Zeros in a Given | 2-Majority Element: Attempt | 3-Finding Floor Value: Attem | 4-Two Elements sum to x: Att | 5-Implementation of Quick S
File C:/Users/Charulatha/Downloads/5-Implementation%20of%20Quick%20Sort_%20Attempt%20review%20(1).mht
Chat

Started on	Saturday, 5 September 2026, 8:26 AM
State	Finished
Completed on	Saturday, 5 September 2026, 8:33 AM
Time taken	7 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Write a Program to Implement the Quick Sort Algorithm

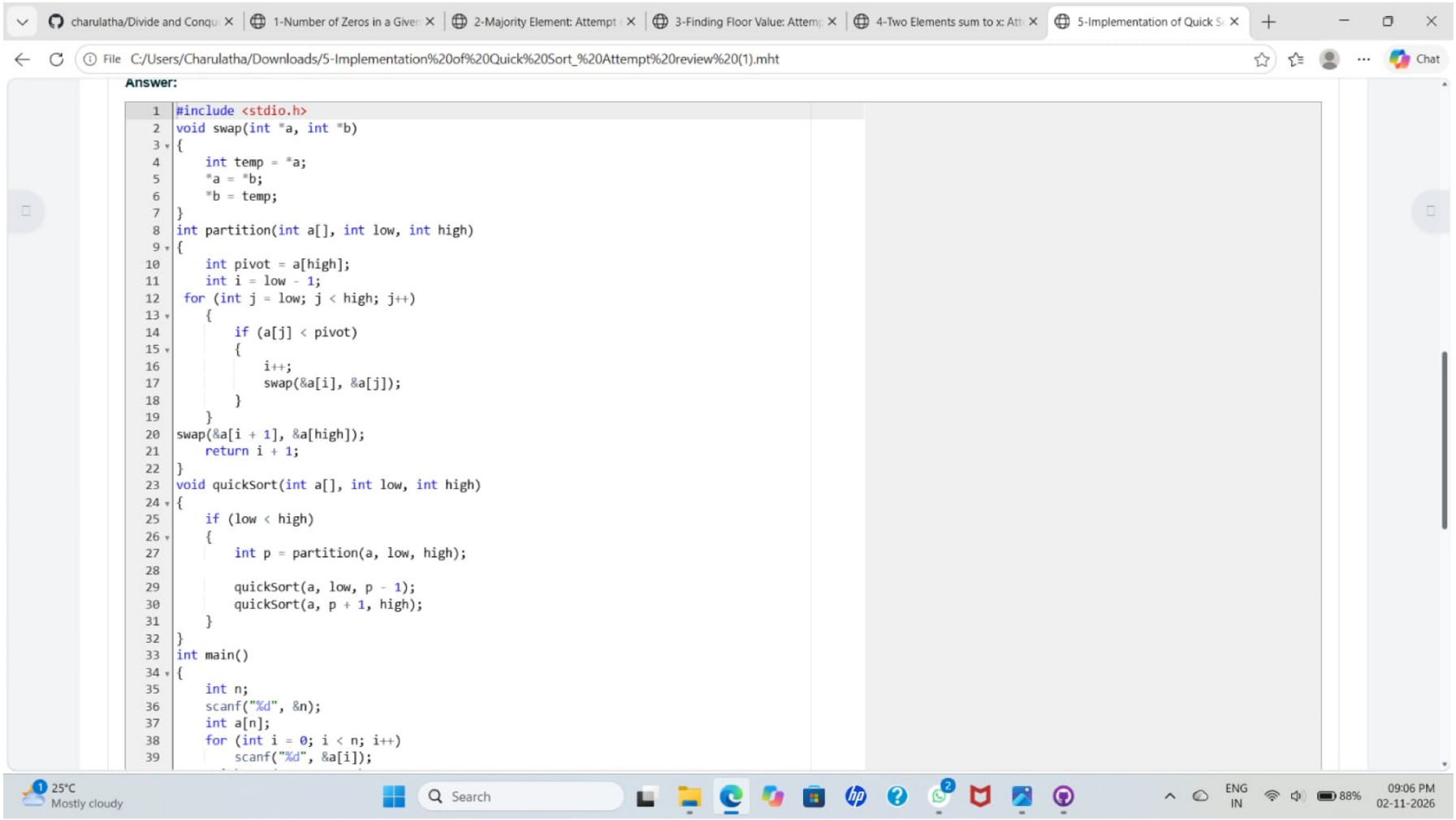
Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

25°C Mostly cloudy
Search
ENG IN
88%
09:06 PM 02-11-2026


charulatha/Divide and Conqu
1-Number of Zeros in a Given
2-Majority Element: Attempt
3-Finding Floor Value: Attem
4-Two Elements sum to x: Att
5-Implementation of Quick S
File C:/Users/Charulatha/Downloads/5-Implementation%20of%20Quick%20Sort_%20Attempt%20review%20(1).mht
Chat
Answer:
#include <stdio.h>
void swap(int *a, int *b)
{
int temp = *a;
*a = *b;
*b = temp;
}
int partition(int a[], int low, int high)
{
int pivot = a[high];
int i = low - 1;
for (int j = low; j < high; j++)
{
if (a[j] < pivot)
{
i++;
swap(&a[i], &a[j]);
}
}
swap(&a[i + 1], &a[high]);
return i + 1;
}
void quickSort(int a[], int low, int high)
{
if (low < high)
{
int p = partition(a, low, high);
quickSort(a, low, p - 1);
quickSort(a, p + 1, high);
}
}
int main()
{
int n;
scanf("%d", &n);
int a[n];
for (int i = 0; i < n; i++)
scanf("%d", &a[i]);
25°C
Mostly cloudy
Search
ENG
IN
88%
09:06 PM
02-11-2026

charulatha/Divide and Conqu | 1-Number of Zeros in a Given | 2-Majority Element: Attempt | 3-Finding Floor Value: Attem | 4-Two Elements sum to x: Att | 5-Implementation of Quick S

File C:/Users/Charulatha/Downloads/5-Implementation%20of%20Quick%20Sort_%20Attempt%20review%20(1).mht

```
        scanf("%d", &a[i]);
    quickSort(a, 0, n - 1);

    for (int i = 0; i < n; i++)
    {
        printf("%d", a[i]);

        if (i < n - 1)
            printf(" ");
    }

    return 0;
}
```

	Input	Expected	Got	
☐	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	☐
☐	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	☐
☐	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.

Finish review

25°C Mostly cloudy | Search | ENG IN | 88% | 09:06 PM 02-11-2026